Defect-Aware Multi-Die Image Fusion for Semiconductor Reference Reconstruction

Md Shamiul Islam Riyad · B.Sc. in CSE, United International University

October 2026



Fig. 1. Proposed workflow (conceptual design; each stage is to be developed and tested in this project).

1 Problem statement and motivation

Wafer inspection usually judges a die by comparing it with a reference, such as a neighbouring die (die-to-die) or a design rendering. If the reference is a single inspection image, its own defects, noise or imaging artefacts enter the comparison and can cause missed or false detections; a verified defect-free "golden" image is not always available for every layer and imaging condition. Dies of the same design share one intended pattern, while random defects usually occur at different positions on different dies, so a small aligned group holds complementary evidence: at most locations, most dies are clean. This project asks whether that evidence can be fused into a *candidate reference image* (a digital estimate of the defect-free pattern for inspection, not a physically fabricated or repaired die) while marking regions that cannot be inferred reliably.

2 Research gap

Each building block is well studied: supervised defect classification [4], unsupervised anomaly detection and segmentation learned from defect-free samples [1, 2], image registration, and multi-image fusion. The open question is their combination when *every* available image may be defective: per-pixel reliability estimated jointly across 4–5 same-design dies, fusion of only trustworthy evidence, and an explicit uncertainty output. No novelty is claimed until a systematic literature and patent review, the first work package, is complete.

3 Research objectives

- **O1** Register groups of 4–5 images and estimate reliability masks; report alignment error (pixels) on known synthetic transforms.
- **O2** Build confidence-weighted fusion that outputs a candidate reference and a per-pixel uncertainty map; compare with single-image, mean, median and anomaly-masked-mean baselines.
- **O3** Quantify fidelity and defect leakage while varying defect count, size, inter-die overlap, noise and misalignment.
- **O4** Test whether uncertainty tracks reconstruction error, and release a reproducible evaluation pipeline.

4 Proposed methodology (Fig. 1)

Registration: intensity normalisation, global alignment (e.g. phase correlation), then local refinement; alignment residuals lower a pixel's weight. **Reliability:** per-pixel weights from an anomaly detector trained on defect-free data, deviation from a robust cross-die median, and registration confidence. **Fusion:** a weighted robust estimator (e.g. weighted median); a learned fusion network is an optional extension. **Uncertainty:** from the number of reliable observations and their weighted spread; pixels below a support threshold are flagged, not inpainted, to avoid invented structure. Systematic defects repeated on every die cannot be removed this way and are treated as a known failure mode.

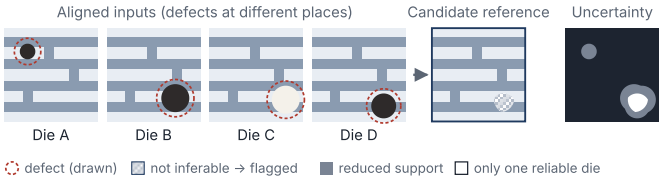


Fig. 2. *Illustrative schematic drawn by hand, not experimental data.* Where the defects in B, C and D overlap, only die A is reliable, so the region is flagged rather than guessed.

5 Expected contributions

(a) A prototype of defect-aware multi-die fusion with an uncertainty output; (b) a reproducible evaluation pipeline (synthetic-group generator, fixed splits, configurations, scripts); (c) quantitative comparison with the O2 baselines and documented failure modes. No performance outcome is assumed in advance.

6 Dataset feasibility

Dataset	Content (per provider documentation)	Groups? / Clean ref.?
Carinthia; Carinthia-S [3]	4,591 grayscale SEM defect images, one production layer of unstructured wafers, 6 classes; Carinthia-S adds binary masks	No / No
MVTec AD [1]	5,354 images, 15 object and texture categories; defect-free training set; pixel-precise anomaly masks	No / No
MVTec AD 2 [2]	>8,000 images, 8 scenarios incl. lighting changes; private-test ground truth withheld	No / No

None provides same-design image groups with complementary defects and a clean ground-truth reference (MVTec defect-free images are not paired with defective ones). The project will therefore (1) build **controlled synthetic groups**: replicate a clean pattern image N times with small shifts, noise and intensity drift, then paste real defect appearances (cut out via Carinthia-S or MVTec masks) at known positions, giving exact ground truth; and (2) seek **industry collaboration** for real multi-die SEM or optical sets with golden or design-rendered references. Without (2), conclusions are limited to synthetic data. MVTec data are CC BY-NC-SA 4.0 (non-commercial).

7 Evaluation

- **Image similarity:** PSNR, SSIM against the clean reference.
- **Structure:** edge F-score; line-width and edge-placement error.
- **Defect leakage:** inserted-defect pixels still above an error threshold; downstream detection effect (AUROC, AU-PRO).
- **Registration:** mean/max displacement error (px) vs. known transforms.
- **Uncertainty:** rank correlation with error, sparsification error, recall of unrecoverable regions.

Ablations: number of dies (2–5), inter-die defect overlap, noise level.

8 Selected references

[1] P. Bergmann, M. Fauser, D. Sattlegger, C. Steger, "MVTec AD — A comprehensive real-world dataset for unsupervised anomaly detection," *CVPR*, 2019, pp. 9584–9592. doi:10.1109/CVPR.2019.00982 · dataset

[2] L. Heckler-Kram, J.-H. Neudeck, U. Scheler, R. König, C. Steger, "The MVTec AD 2 dataset: Advanced scenarios for unsupervised anomaly detection," arXiv:2503.21622, 2025. · dataset

[3] C. Kofler *et al.*, Carinthia SEM defect dataset, Zenodo, 2024, doi:10.5281/zenodo.10715190; C. Kofler, V. Hasić, Carinthia-S (with segmentation masks), Zenodo, 2025, doi:10.5281/zenodo.16895427.

[4] K. Taha, "Observational and experimental insights into machine learning-based defect classification in wafers," *Journal of Intelligent Manufacturing*, Springer. doi:10.1007/s10845-024-02521-0